

Exercise 05^a: Check the idempotence of openings for a particular case

To verify the idempotence of the opening operator, we used the image `cam_74.pgm` with a size of 1. We executed the program `exercise_04a_opening.py` that generated this image:



Then we used the same opening to generate the second image:



Afterwards, we used the code from `exercise_02b_compare.py` to compare both images. The comparison result was 1, proving that both images are identical. Therefore, we can confirm the idempotence property of the opening operator, meaning that once an opening operator is applied, if we use the opening operator once again, the image will not be modified:

$$\textit{Opening}(\textit{Opening}(I)) = \textit{Opening}(I)$$

So, the results obtained validate that the opening operator behaves as the expected properties of mathematical morphology.

Exercise 05b: Idem for closings

To verify the idempotence property of the closing operator, we used the image `cam_74.pgm` with a size of 1. We executed the program `exercise_04a_closing.py` that generated this image:



Then we used the same closing to generate the second image:



Afterwards, we used the code from `exercise_02b_compare.py` to compare both images. The comparison result was 1, proving that both images are identical. Therefore, we can confirm the idempotence property of the closing operator, meaning that once an closing operator is applied, if we use the closing operator once again, the image will not be modified:

$$Closing(Closing(I)) = Closing(I)$$

So, the results obtained validate that the closing operator behaves as the expected properties of mathematical morphology.